

SafeSpeed

Is the posted speed limit survivable? A Safe System screening of the road networks of Maharashtra and Thailand

Methodology report · AI for Safer Roads 2026 Innovation Challenge (Asian Development Bank / Agilysis) · results built 2026-09-28

Key terms

Term	Meaning	Term	Meaning
v50, v85	median and 85th-percentile speed of traffic on a road (km/h)	Band	how soon to act on a segment: Aligned, Low, Moderate, High, Critical
Safe speed	highest speed at which the crash a road allows is survivable for its most exposed user	iRAP, ThaiRAP	International Road Assessment Programme and its Thai arm; their surveyors code road attributes from images
Headline	the SafeSpeed score and safe speed; the other models run beside them	POI; VRU	point of interest (school, stop, shop ...); vulnerable road user: people on foot, cyclists, motorcyclists
KSI	killed or seriously injured (MAIS3+F: Abbreviated Injury Scale 3 or more, or death)	GBM	gradient-boosted decision trees
Expected harm	chance that a crash kills or seriously injures the most exposed user, averaged over the road's speeds	Graph neural network	a model that also reads the features of connected road pieces (here GraphSAGE)
Design speed	the speed a road's layout and surroundings invite, predicted from context alone	Stack	weighted sum of several models' predictions; weights non-negative, fitted out of sample
Segment; road piece	one of the 15,554 road sections we score; one of the 46,548 shorter TomTom pieces they are made of	R²; κ; ρ	share of variance explained (1 = perfect); Cohen's kappa, agreement beyond chance (0 = chance); Spearman rank correlation
Probe	a TomTom probe vehicle (probe passes: its measurements), or a small classifier on an image embedding	pp	percentage points
CLIP, DINOv3	pretrained image models; an embedding is the vector they make from a photo	Spatial CV	testing on 50 km blocks of road that the model never saw in training

Running the deliverables

The code of the three models, their examples and expected results and this report are on [GitHub](#). Their weights and sample inputs are one download, SafeSpeed_data_weights.zip, from Google Drive at <https://drive.google.com/drive/folders/lj-Lf5c8Cs0SkGGhmWzDX-TC30KZAR8ni>. Everything then runs on a laptop CPU with Python 3.9 or later, numpy and pandas, with no GPU or internet connection.

```
git clone https://github.com/Shishuii/safe_speed.git
cd safe_speed
# unzip SafeSpeed_data_weights.zip here: data/ and weights/ sit next to run_inference.py
python3 -m venv .venv
source .venv/bin/activate      # Windows: .venv\Scripts\activate
python -m pip install -r requirements.txt
python run_inference.py --check
python run_vision.py --check
python run_ensemble.py --check
```

Command	What it runs	Writes to outputs/
run_inference.py	Design-speed model: invited v50 and v85, each factor group's share, expected harm. Sample: 200 segments from both countries.	predictions.csv
run_vision.py	Vision model, Thailand: image attributes, safe speed, invited speed, harm; --with-dinov3 adds DINOv3 classifier scores. Sample: 3 segments, 7 street images.	vision_predictions.csv, vision_images.csv
run_ensemble.py	Research ensemble: v85 from context, road graph and images; seven members by default, --stack with_dinov3 adds the DINOv3 member, --stack without_images keeps four. Sample: 9 road pieces of 5 segments.	ensemble_predictions.csv

With --check, each run compares its outputs with the pipeline's and ends with check: PASSED. Three examples in examples/ call the models from Python. run_inference.py --input my_roads.csv scores your roads (columns: COLUMNS.md). New street images are embedded with run_vision.py --embed-to NEW.npz, which needs pip install -r requirements-vision.txt and a one-off CLIP download (about 1.7 GB); --with-dinov3 adds the DINOv3 backbone (about 1.2 GB). Image models and outputs are non-commercial (Section 9).

1. The problem

How badly people are hurt in a crash depends on how fast the vehicles are moving. Limits are usually set by road class and habit, and drivers settle at the speed the road's layout invites. Neither is checked against the human body. On the injury curve we use, the chance that a person on foot struck at 30 km/h is killed or seriously injured is 11 %; at 50 km/h it is 36 %.

The Safe System approach (Tingvall & Haworth 1999; WHO 2017; OECD/ITF 2018) starts from the fact that people make mistakes. Crashes will happen, so the road has to keep their forces within what a body can survive. For speed this becomes a plain test. Where people walk beside or across traffic, cars must be slow enough for a pedestrian to survive being hit. Where two cars can meet head-on, they must be slow enough for the occupants to survive. The survivable speed depends on who can be hit and how.

India and Thailand carry a heavy share of the world's road deaths. WHO estimates 216,618 road deaths in India in 2021 (15.4 per 100,000 people) and 18,218 in Thailand (25.4 per 100,000). Riders of motorcycles and other two- and three-wheelers account for 45 % of reported deaths in India, and pedestrians for another 19 %. In Thailand two- and three-wheeler riders are 51 % of reported deaths, with 46 % recorded as other or unknown, and these vehicles make up 52 % of all registered vehicles. The national maximum urban limit is 70 km/h in India and 80 km/h in Thailand, far above the speed a pedestrian or a motorcyclist survives (WHO 2023).

The challenge gave us the road networks of Maharashtra and Thailand with posted limits and TomTom probe speeds: the median, the 85th percentile and a probe count for each section. Of 69,966 sections, 15,554 have a valid analysis status; these are the segments we score (4,010 in Maharashtra, 11,544 in Thailand). We added the sources in Table 1.

Neither network comes with crash records, so no model here can learn where crashes happen or how often. We screen for severity instead: if a crash happens on this road at the speeds measured, how likely is death or serious injury?

A road authority needs a list it can act on. For every segment the tool therefore says what speed the road can safely carry, how far the posted limit sits above it, who is exposed, how soon to act and what to do first. Each score splits into named parts, and every result in this report is read from the build outputs when the report is generated.

Some things are out of scope. We do not predict crash counts or rates, and we do not compute benefit-cost ratios. Enforcement and driver behaviour are not modelled. The speed models do not transfer between the two countries (R^2 -0.321 from Maharashtra to Thailand, -0.123 the other way), so each country has its own. Roads outside the two networks are not covered.

Dataset	What it gives	Used for
Challenge road network (ADB / TomTom)	network, posted limit, v50, v85, probe count per section	labels, limit gap, traffic volume
TomTom granular Traffic Stats	19 speed percentiles on 46,548 road pieces	speed distribution, measured harm, research labels
Challenge feature files	counts of schools, stops, shops and other POIs; observed helmet use	exposure; model features
WorldPop 2025, 100 m	residents within 300 m of the road	exposure; model features
GHS-SMOD R2023A, 1 km	settlement class, urban centre to rural	safe speed only
OpenStreetMap	lane count, one-way	safe speed (lane count); labels for two image probes
city2graph	graph measures and road graphs	research models
Mapillary	street images and hosted object detections, mainly Thailand	image models; checks on actions
V-RoAst / ThaiRAP	Thai survey images with iRAP codes by ThaiRAP staff	training the image models
Published research	injury curves, iRAP Toolkit, Power Model	safe speed, harm, actions

Table 1. Data sources. Full citations are in the references.

2. How SafeSpeed works

Figure 1 follows one segment through the method, from the data to the recommended action. The vision model and the research ensemble run beside the score and never change it.

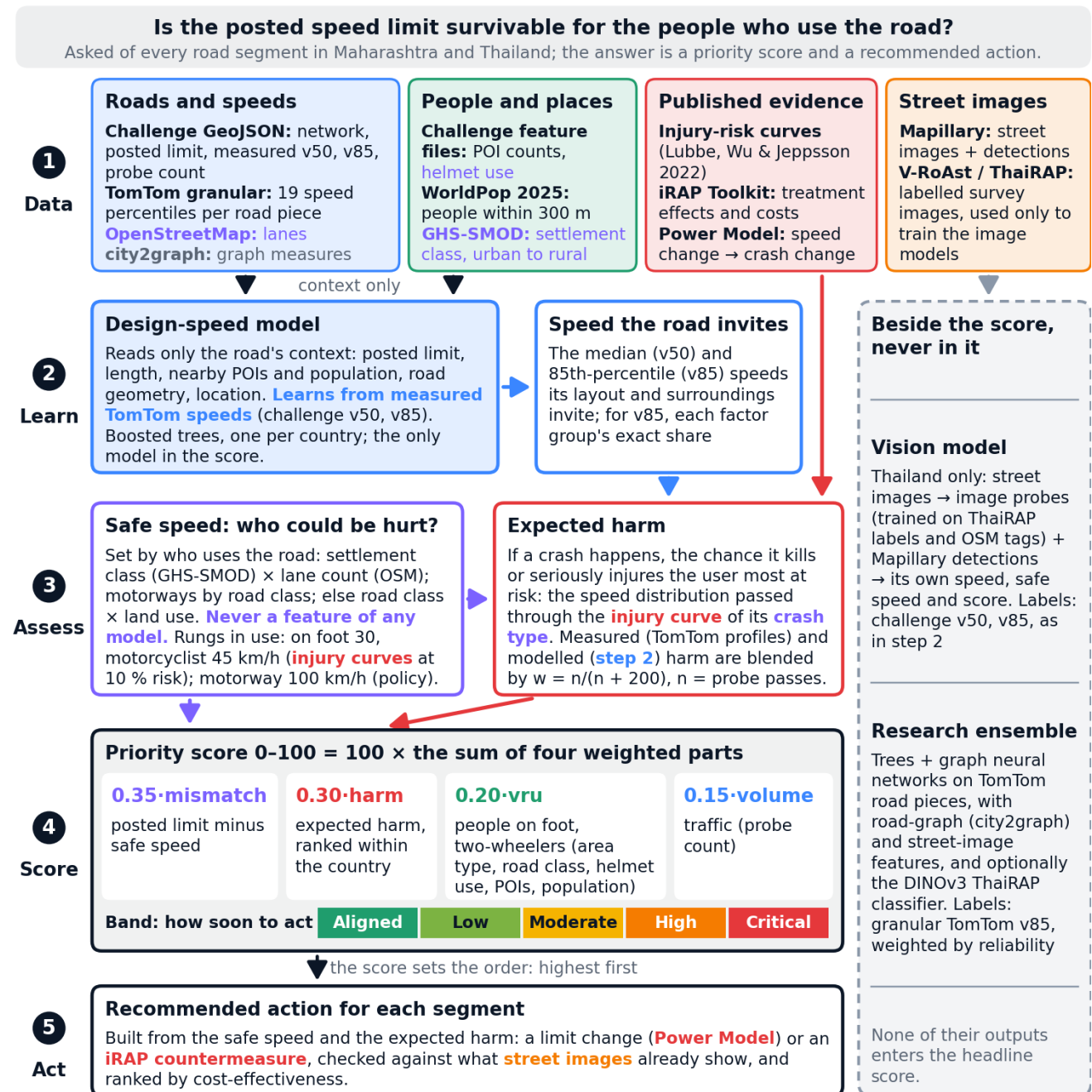


Figure 1. The method in five steps. Purple inputs are never a feature of any model; settlement class and lanes set the safe speed. Coloured words tie a step to the data it uses. The models in the dashed panel run beside the score.

3. Labels: measured speed

Our models learn the one measured target we have, speed: log v50 and log v85, the speeds a road's design and surroundings invite. Harm is computed from them afterwards (Figure 2). From TomTom's 19 speed percentiles per road piece we rebuild the full speed distribution (monotone interpolation on the probit scale, Gaussian tails); its mean and standard deviation are off from TomTom's own by 0.14 km/h and 0.56 km/h on average over 1,479 road pieces.

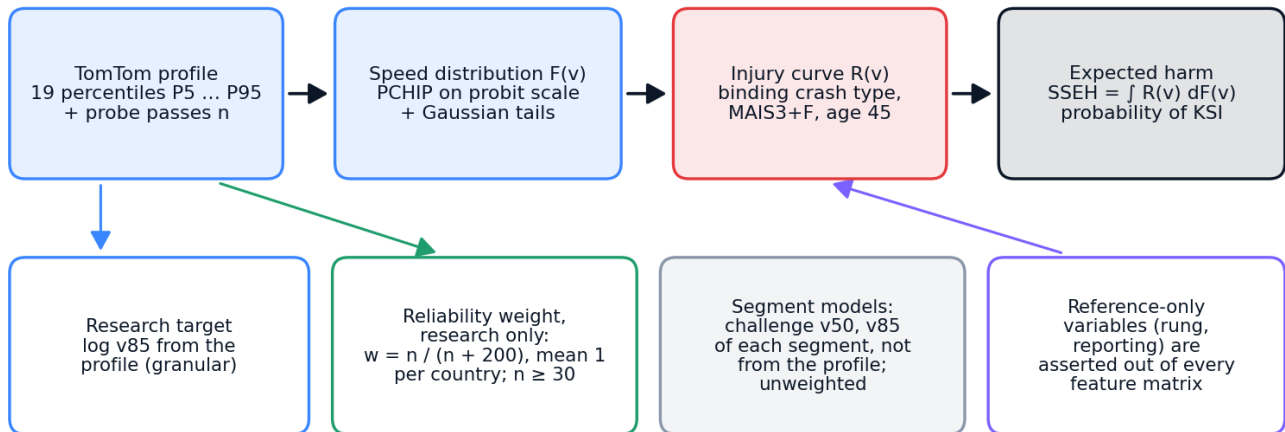


Figure 2. From speed profile to expected harm and labels. The research models learn log v85 from the profile, weighted by reliability; the segment models learn the challenge v50 and v85. The variables that choose the crash type never enter a model.

We do not learn harm directly. A harm label is built from the same inputs that choose the crash type (settlement class, lane count), so a model can score well by re-learning that rule. We measure this with a circularity floor, the R^2 reachable from the rule inputs alone, and judge each target by its skill above it. Log v85 has skill 0.336, harm for one crash type 0.158 and harm weighted by the road-user mix -0.244 (Figure 3a). In a wider label study, the label with the highest R^2 ranked fifth on skill (Figure 3b).

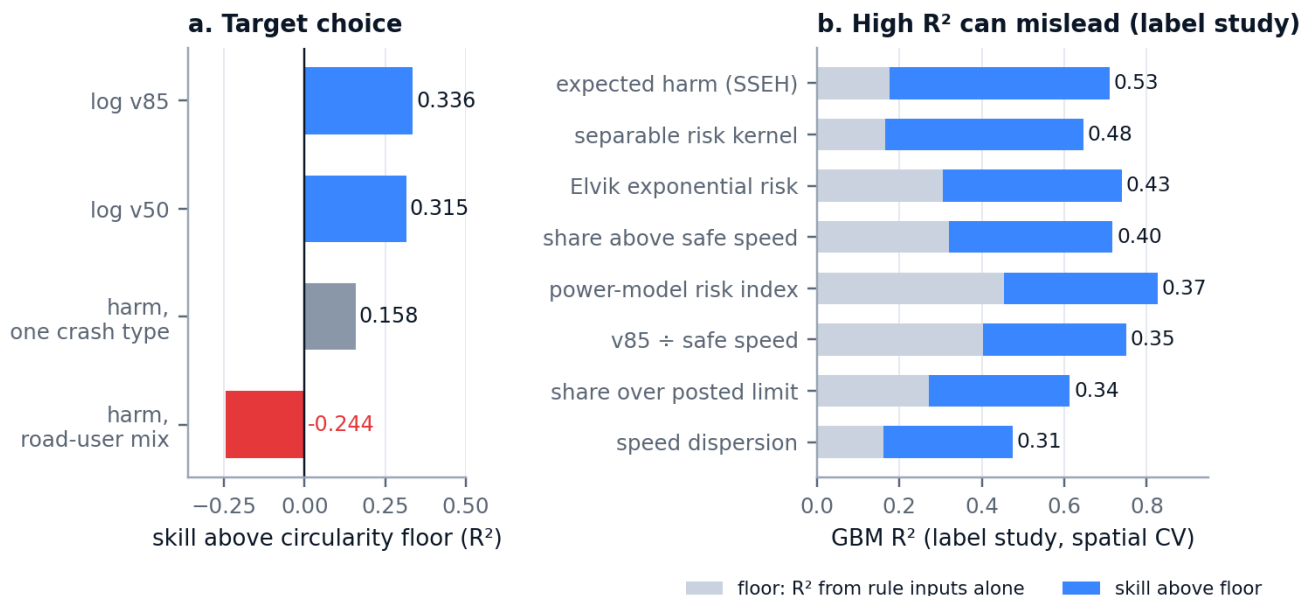


Figure 3. Skill above the circularity floor. a: the target of the design-speed model. b: a separate label study on the road pieces; the grey part of each bar is what the rule inputs alone explain. Its harm label (SSEH) uses fatality curves and takes the crash type from road class and land use only, so it is not the harm target of panel a.

For the same reason, the 22 reference-only columns, which build the safe speed (settlement class, OSM lanes and one-way, the challenge's road class and land use) or describe who uses the road (helmet use, two-wheeler share, road-user mix), never enter a model (tests check this). They would add 0.058 R^2 to the design-speed model. Speed models may read TomTom's road class and lanes or land use seen in images, since they predict speed, not harm.

Neighbouring roads share speeds, so a random split flatters a model. Each speed model is tested on whole 50 km blocks it never saw; a random split would add +0.022 R^2 to the design-speed model (Figure 4).

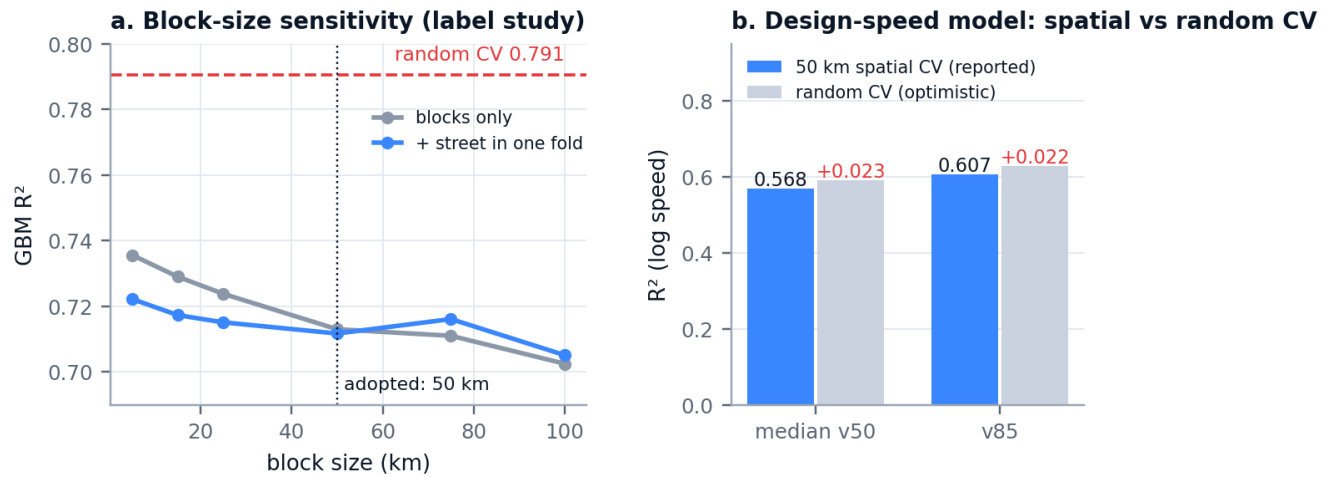


Figure 4. a: R^2 against block size in the label study; the dashed line is a random split. b: the design-speed model under spatial and random cross-validation.

4. Models

4.1 Design-speed model

This is the only model inside the score. For each country, and for log v50 and log v85, we fit gradient-boosted trees (depth 2, 1,000 trees, learning rate 0.06) on the 15,143 segments with measured speed. The model reads context only: posted limit, length, nearby schools, transport stops and other POIs, population, road geometry and location. Features may interact only inside six factor groups, so each prediction splits exactly into one contribution per group (largest residual 6×10^{-14}). The constraint costs 0.008 R^2 .

Spatial-CV R^2 for log v85 is 0.607 pooled, 0.416 in Maharashtra and 0.512 in Thailand; the mean absolute error is 10.6 km/h. The pooled value also counts the difference between the two countries, so the per-country values are the fairer measure.

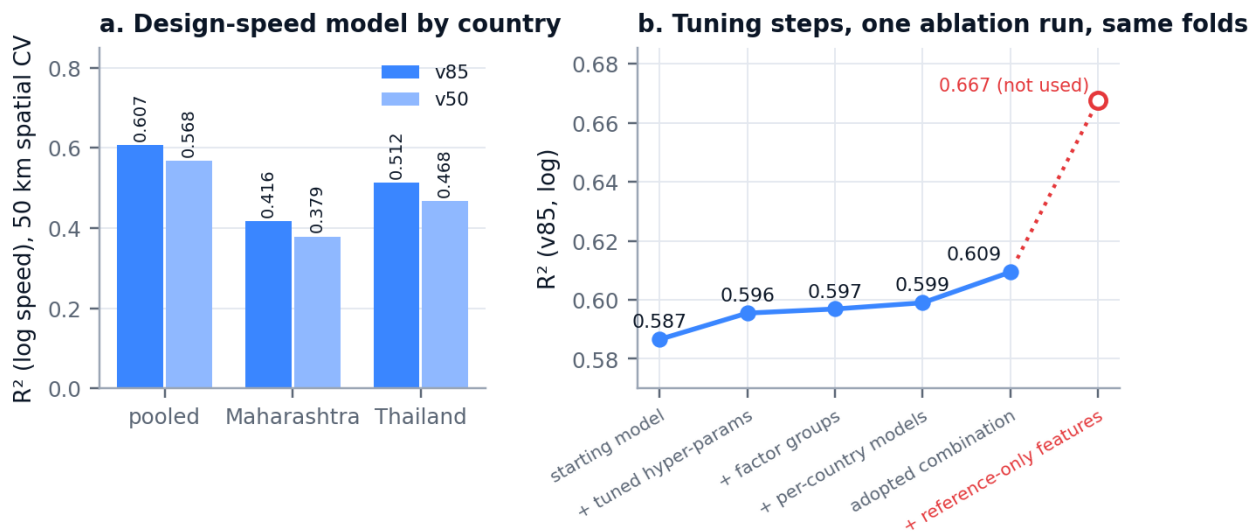


Figure 5. a: R^2 by country and target. b: one ablation run on the same folds, from the starting model to the adopted one. Adding the variables that set the safe speed would score higher and is not done.

4.2 Vision model: CLIP probes and Mapillary detections

The vision model asks how much of the speed model can be rebuilt from street images alone. It covers Thailand, where Mapillary imagery is dense. We embed 45,294 images that face along the road with a frozen CLIP ViT-L/14 (Radford et al. 2021). Linear probes trained on ThaiRAP survey labels read 11 iRAP attributes, such as lighting, shoulders, roadside hazards and traffic calming. An attribute is kept only if it reaches $\kappa \geq 0.20$ and beats the majority class on held-out ThaiRAP segments. Two probes trained on OpenStreetMap tags read the lane count and whether the road is divided, and Mapillary's own detections show people, riders, footpaths and crossings. The first 32 principal components of the embedding describe overall appearance (Figure 6).

ThaiRAP labels (V-RoAst) train the probes; each attribute must reach $\kappa \geq 0.20$ and beat the majority class

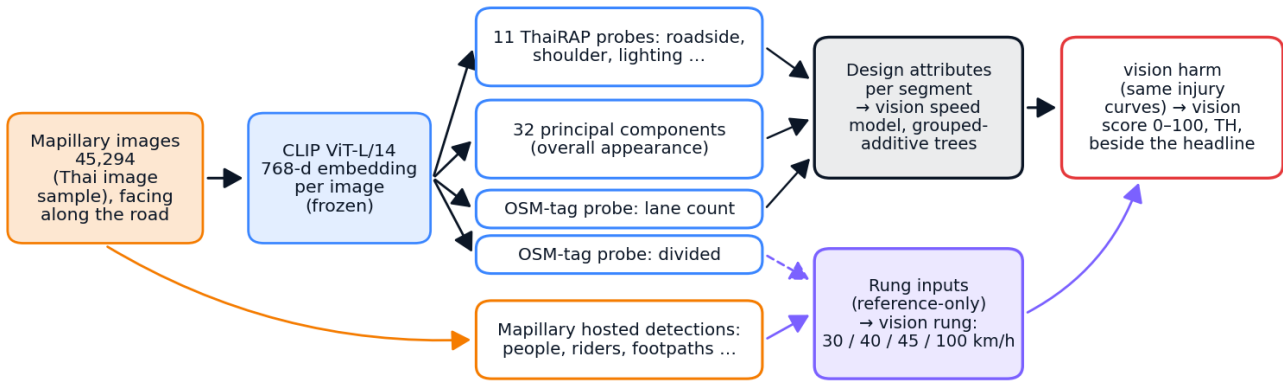


Figure 6. The vision model. Probes and embedding components feed the speed model; detections and the divided-road probe set the vision safe speed.

On 6,268 Thai segments the images alone reach v85 R^2 0.437, against 0.523 for the context model on the same rows and 0.569 for both together (Figure 7a). The vision safe speed agrees with the headline one on 61 % of segments (Figure 7b). Where they differ, the segment is a candidate for a field check; neither is ground truth.

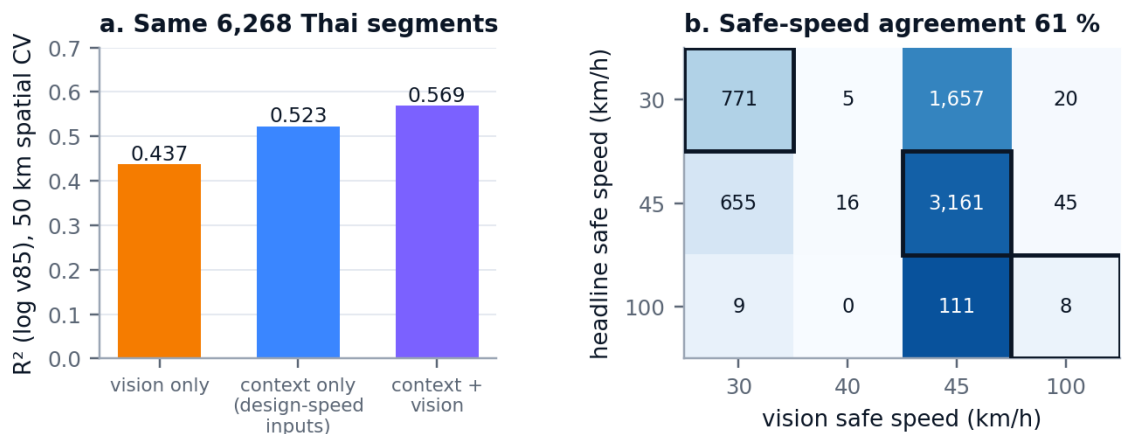


Figure 7. a: images, context and both, on the same Thai segments. b: headline against vision safe speed (segment counts); outlined cells agree.

We also tried coding iRAP attributes by asking vision-language models directly, the method of V-RoAst (Jongwiriyanurak et al. 2025). On ThaiRAP's labels it fails: mean κ 0.11 for Gemini-1.5-flash, 0.05 for GPT-4o-mini and 0.01 for a local Qwen2.5-VL-3B (Figure 8a). On the attributes kept, the CLIP probes reach κ 0.43, against 0.14 and 0.05 for the paper's two commercial models on the same segments.

4.3 DINOv3 classifier trained on ThaiRAP

Our second image model also puts one linear head per attribute on a frozen backbone, but the backbone is DINOv3 ViT-L/16 (Siméoni et al. 2025) and each image's features are pooled five ways: the global token, the mean patch, and the left, centre and right thirds of the image. The side pools matter because iRAP codes many attributes separately for the driver and passenger side; Thailand drives on the left, so the passenger side is the image's left. Each training image is also used mirrored, with its driver- and passenger-side labels swapped, which doubles the data for side attributes. The heads cover 30 ThaiRAP attributes with 109 classes, against the 11 read by the CLIP probes, and are trained on V-RoAst's ThaiRAP labels.

For testing we held out every fifth 2 km block of the survey route, so neighbouring images never fall on both sides of the split: 1,573 training images and 464 test images on 119 segments. Over the 25 attributes that vary on those segments, mean κ is 0.34, against 0.11 and 0.04 for the paper's Gemini-1.5-flash and GPT-4o-mini outputs on the same segments (Figure 8a). Of these, 16 pass the same gate as the probes: motorcycle flow, area type, median type, roadside object (both sides), roadside distance (passenger side), paved shoulder (both sides), property access, number of lanes, curvature, skid resistance, pedestrian crossings, traffic calming, sidewalk (passenger side) and service road.

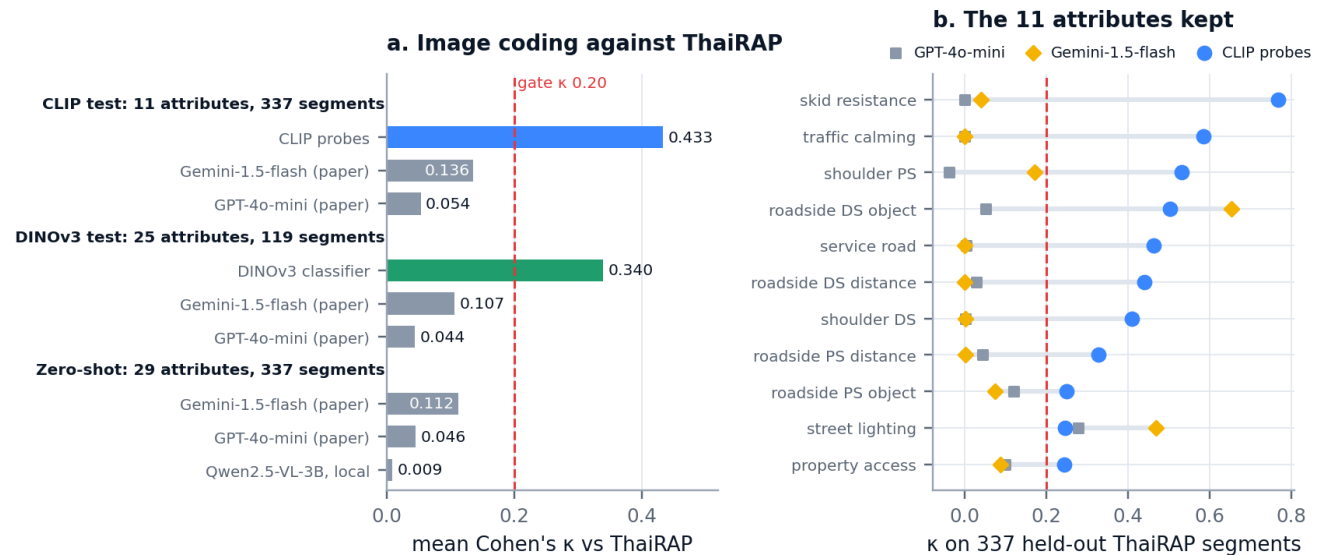


Figure 8. a: mean Cohen's κ against ThaiRAP labels; each group compares models on the same attributes and segments. b: per attribute, the CLIP probes against the paper's VLM outputs (PS/DS = passenger/driver side).

The heads are run on every usable Thai street image, and each segment gets the mean class probability per attribute. Only attributes with a design role may enter a model. Those that describe who uses the road or its junctions, such as area type, median type and sidewalks, would re-learn the safe-speed rule and stay out (Section 3); lane count and land use read from the images do enter, as in the vision model. The 70 class probabilities of its 19 design-role attributes feed an optional eighth member of the research ensemble (--stack_with_dinov3; Section 4.4). All 19 enter, including 8 that fail the κ gate (11 pass); the trees decide how much to use them.

4.4 Research ensemble

The research ensemble tests how well speed can be predicted with every signal we have, on the 46,548 road pieces with at least 30 probe passes. Rows are weighted by reliability, $w = n/(n + 200)$, and the five folds are 50 km blocks with each named street and carriageway pair kept whole. Nothing on the scoring path reads it.

Gradient-boosted trees read 15 context features, or 28 with 13 city2graph measures of the road network. Two GraphSAGE networks (Hamilton et al. 2017) also read their neighbours' features, one on the graph of shared endpoints and one on a 12-nearest-neighbour graph (Figure 10). Image members add the vision model's 43 street-image columns to the trees and to both networks. The DINOv3 member adds the classifier's 70 class probabilities to the image tree member (141 features). A non-negative least-squares stack, fitted within each fold, combines the members.

Without images the four context members reach R^2 0.665 (Maharashtra 0.491, Thailand 0.641; Figure 9a). city2graph adds little: +0.014 R^2 over the base trees (Figure 9a), and +0.010 in an unweighted ablation, less than latitude and longitude (+0.024; Figure 9b).

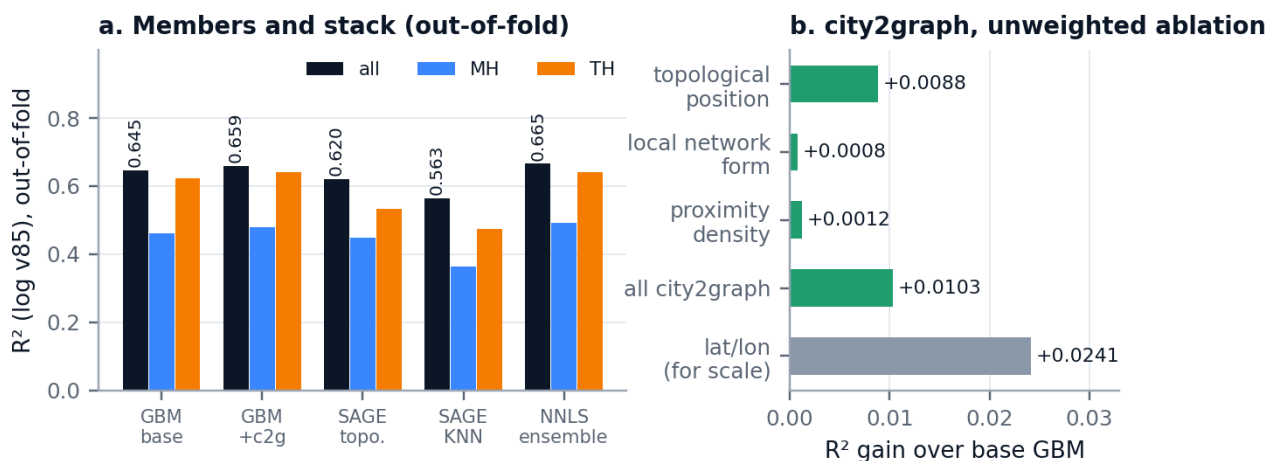


Figure 9. a: out-of-fold R^2 of the context members and their stack, overall and by country. b: gain of each group of city2graph features over the base trees, with coordinates for scale.

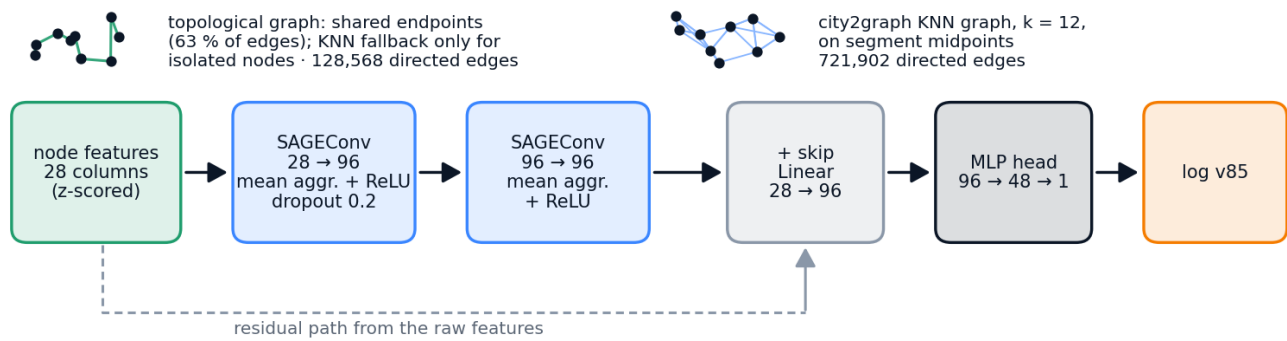


Figure 10. The GraphSAGE member: two mean-aggregation layers, a skip path from the raw features and a small head. The same network runs on both graphs.

Images help where they exist. On the Thai road pieces with imagery, the CLIP image members lift the stack from 0.638 to 0.672 (+0.034, 95 % block-bootstrap interval +0.022 to +0.044). Shuffling the image values among segments removes the gain (0.637). The gain sits in the tree members. The graph networks are unstable: storing the same image inputs at a different numeric precision changes their R^2 by up to 0.024, so we do not credit their changes to the images (Figure 11). Outside the imagery the stacks score the same (+0.004 on Thai road pieces without images, −0.001 in Maharashtra).

The DINOv3 member adds little on top. On the same Thai rows it moves the stack from 0.672 to 0.673 (+0.0010, interval +0.0003 to +0.0017); the tree member with DINOv3 scores is no better than the same member without them (difference +0.0000, interval −0.0066 to +0.0050). With the scores shuffled among segments the stack reaches 0.674, as high or higher, so none of the gain is credited to what the classifier reads. The package runs the seven-member stack by default; `--stack with_dinov3` adds the DINOv3 member.

Stack	Members	All	MH	TH	TH with images	TH without images
without_images	4	0.665	0.491	0.641	0.638	0.640
default	7	0.669	0.490	0.661	0.672	0.644
with_dinov3	8	0.669	0.489	0.661	0.673	0.644

Table 2. Out-of-fold R^2 (log v85) of the ensemble's stacks, named as in the package, which runs default unless given `--stack`.

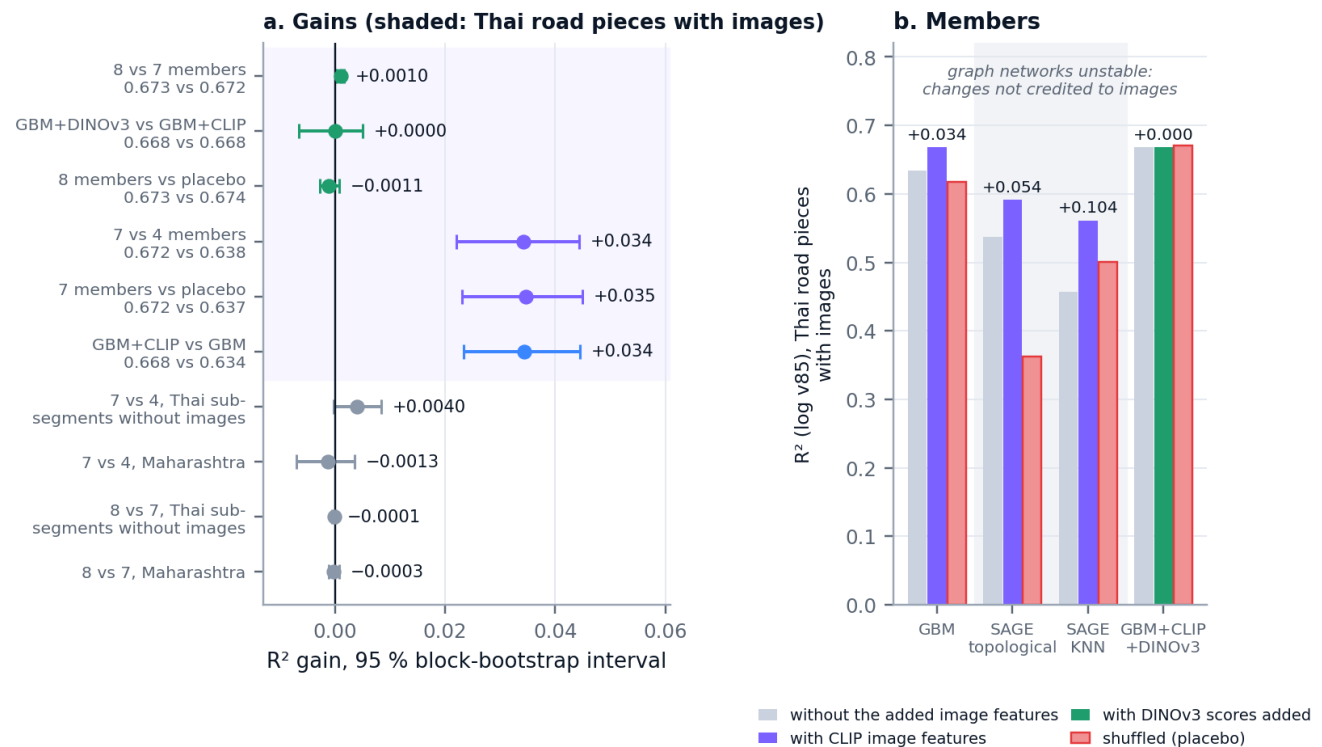


Figure 11. a: R^2 gains with 95 % block-bootstrap intervals; shaded rows are Thai road pieces with imagery. b: members on the same rows without the added image features, with them, and with them shuffled among segments (placebo); in the last group the grey bar is GBM+CLIP and the green bar adds the DINOv3 scores.

5. Safe speed and expected harm

The injury-risk curves come from Lubbe, Wu & Jeppsson (2022), fitted to German in-depth crash data for pedestrians, cyclists, motorcyclists and car occupants struck by the front of a car. We use killed or seriously injured at age 45. In a head-on crash the closing speed is twice the travel speed. The motorcyclist curve rests on 424 cases, 12 of them fatal, and is the least certain.

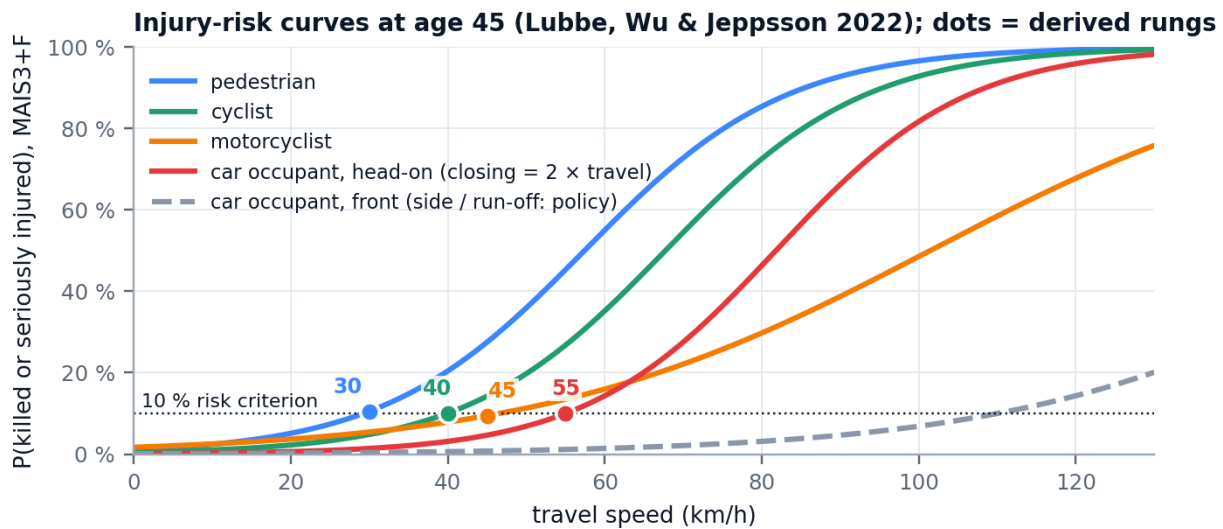


Figure 12. Risk of death or serious injury by travel speed. Dots mark where each curve reaches 10 %, rounded to 5 km/h.

Each rung of the safe-speed ladder is the travel speed at which its curve reaches 10 % risk: pedestrian 30 km/h, cyclist 40 km/h, motorcyclist 45 km/h, head-on 55 km/h. The policy value often quoted for head-on crashes, 70 km/h, carries 27.5 % risk on this curve. Side impact (50 km/h) and motorway (100 km/h) fall outside the source's crash scenario and keep their policy values (Figure 13).

A segment gets the rung of its most vulnerable likely conflict. In dense settlement (GHS-SMOD) people on foot mix with traffic, so the pedestrian rung applies; semi-dense areas get it too unless the road has three or more lanes. Elsewhere we assume motorcycles share the road. Motorways get the motorway rung, and segments without settlement data fall back to road class and land use. The pedestrian rung applies to 5,771 segments, the motorcyclist rung to 9,601 and the motorway rung to 182.

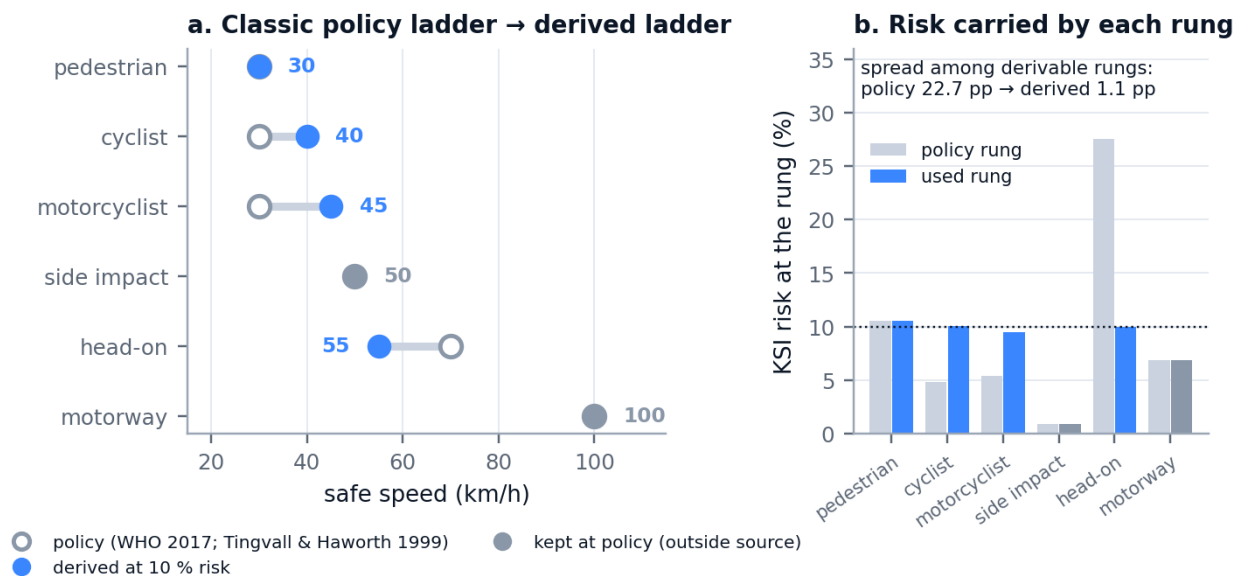


Figure 13. a: the policy ladder (WHO 2017; Tingvall & Haworth 1999) and the ladder derived from the curves. b: the injury risk each rung carries.

Expected harm is the injury risk of the segment's crash type averaged over its speed distribution. Where TomTom profiles cover the road, harm is measured; elsewhere it comes from the design-speed model's v50 and v85 through the same curve. The two are blended by $w = n/(n + 200)$ on probe passes (Figure 14). The median segment has 169,393 passes, so measurement dominates almost everywhere. Harm rebuilt from v50 and v85 alone matches harm from the full profile with R^2 0.946.

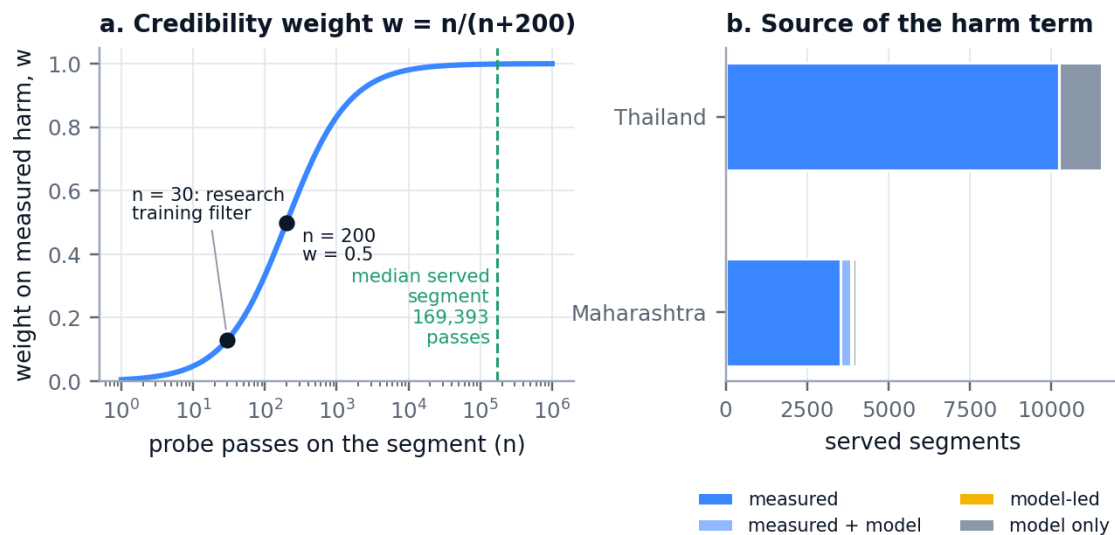


Figure 14. a: weight on measured harm by probe passes. b: where the harm term comes from, by country.

6. Score and recommended action

Each segment's score is a weighted sum: $\text{score} = 100 \times (0.35 \cdot \text{mismatch} + 0.30 \cdot \text{harm} + 0.20 \cdot \text{vru} + 0.15 \cdot \text{volume})$. Each part is scaled to 0–1, and the points are rounded so that they add up to the score exactly. Harm enters as a rank within the country, so it orders roads inside a country but does not compare the two countries (Table 3, Table 4).

Part	Definition	Weight
mismatch	(posted limit – safe speed) / 45 km/h, clipped to 0–1	0.35
harm	within-country percentile of the blended expected harm	0.30
vru	$0.50 \times \text{area and road-class heuristic} + 0.30 \times \text{POIs} + 0.20 \times \text{population}$	0.20
volume	log probe count, scaled between the 5th and 95th percentile	0.15

Table 3. The four parts of the score.

Band	Score	Meaning: how soon to act
Aligned	0–24	Lowest urgency — no action beyond the routine review.
Low	25–44	Low urgency — monitor.
Moderate	45–59	Review the limit at the next routine review.
High	60–77	Prioritise for limit review and engineering.
Critical	78–100	Most urgent — start here.

Table 4. Bands.

Each segment gets one recommended action. Where the limit exceeds the safe speed by more than 5 km/h (14,791 segments), the first option is to lower it to the safe speed. Its effect on fatal and serious crashes (FSI) follows the Power Model (Nilsson 2004; Elvik et al. 2019) on the segment's own speeds: $\text{FSI change} = (v_{\text{after}} / v_{\text{before}})^{3.0} - 1$. We report a realistic case, in which 60 % of the change reaches operating speeds and the effect is capped at 50 %, and a fully enforced case; their median FSI reductions are 50 % and 88 %. Being nearly free, the limit change wins the cost-effectiveness ranking wherever it is offered.

Engineering options come from 14 iRAP Toolkit countermeasures with their published reduction ranges, cost bands and service lives. The best one is reported beside the limit change, because a limit works only where the road supports it. A treatment whose premise is a missing facility is dropped where street images already show that facility (5,137 segments). Actions are ranked by harm reduced per US dollar per km per year.

7. Verification

We check the method against published evidence, independent labels and its own arithmetic (Table 5). The strongest external check is the World Bank's Thailand speed study (2019), which published iRAP-modelled FSI changes for 8 limit changes on two roads. Our Power Model reproduces all of them within 2.5 pp (Figure 15a).

Check	Result	Status
Published evidence and the safe-speed ladder		
World Bank (2019) Thailand case study	8 published FSI changes reproduced; worst gap 2.5 pp	pass
Injury curves vs Lubbe et al. (2022)	worst gap 1.37 pp over 16 published risks	pass
Pedestrian curve vs Rosén & Sander (2009)	largest gap 5.7 pp at 30/50/70 km/h	pass
Policy ladder (WHO 2017) vs derived rungs	risk spread across rungs 22.7 pp (policy) → 1.1 pp (derived)	pass
Image attributes against ThaiRAP		
V-RoAst scorer vs the paper's Table 1	Gemini accuracy 0.818, F1 0.471; GPT-4o-mini 0.751, 0.418, as published	pass
Zero-shot VLM coding	mean κ 0.11 (Gemini), 0.05 (GPT-4o-mini), 0.01 (Qwen2.5-VL-3B)	fail: not used
CLIP probes, same attributes as the VLMs	κ 0.43 vs 0.14 (Gemini), 0.05 (GPT-4o-mini)	pass
DINOv3 classifier, same segments as the VLMs	κ 0.34 vs 0.11 (Gemini), 0.04 (GPT-4o-mini)	pass
Probes on Mapillary images	2 of 5 ThaiRAP-probe outputs reach $\rho \geq 0.20$ with Mapillary detections (the OSM divided-road probe also fails) (Figure 16)	weak
Models and score		
Exact attribution	largest residual 6×10^{-14} ; score points add up exactly	pass
Ensemble reproduction	4 context members retrained; stored predictions matched exactly	pass
Image gain without segments sharing images across folds	+0.033 (interval +0.022 to +0.044), 78 segments dropped	pass
Image gain vs placebo	real over shuffled images +0.035 (interval +0.023 to +0.045)	pass
DINOv3 member vs placebo	real over shuffled scores -0.0011 (interval -0.0027 to $+0.0008$); the package runs the seven-member stack by default and offers <code>--stack with_dinov3</code>	fail: no real gain
Ranking stability	$\rho \geq 0.986$ with the limit-gap weight or all rungs $\pm 20\%$ (Figure 17)	pass
Imagery audit of vulnerable road users	κ 0.16 on 8,065 segments	weak
Transfer between countries	$R^2 -0.321$ (MH→TH), -0.123 (TH→MH)	fail: not claimed

Table 5. Verification checks, including the weak results and failures.

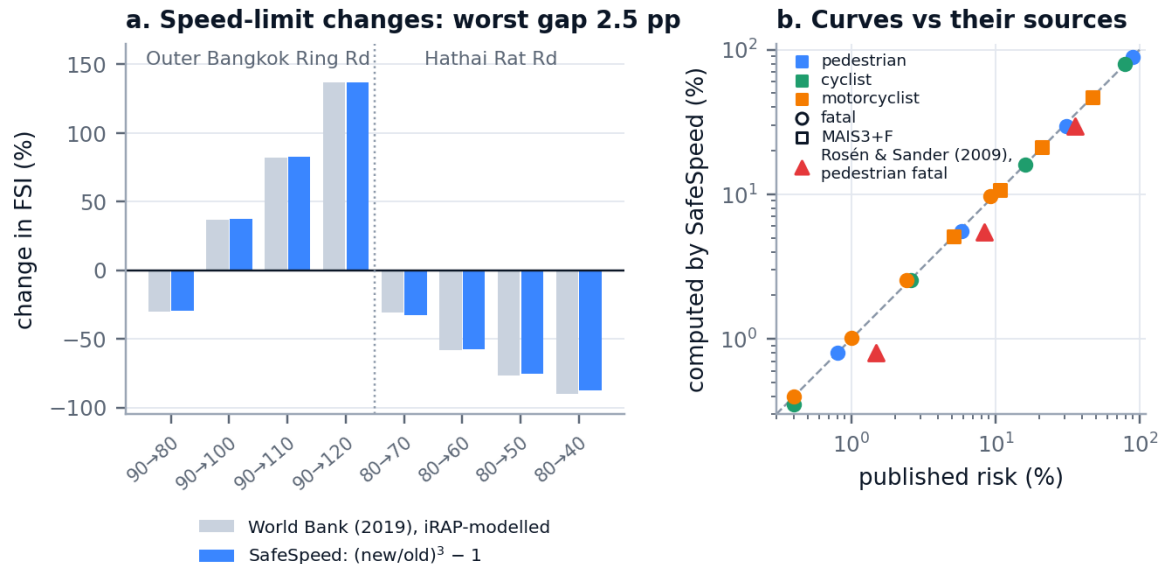


Figure 15. a: FSI change for the World Bank's limit changes, as published and as computed with exponent 3.0. b: published injury risks against the same risks from our curves.

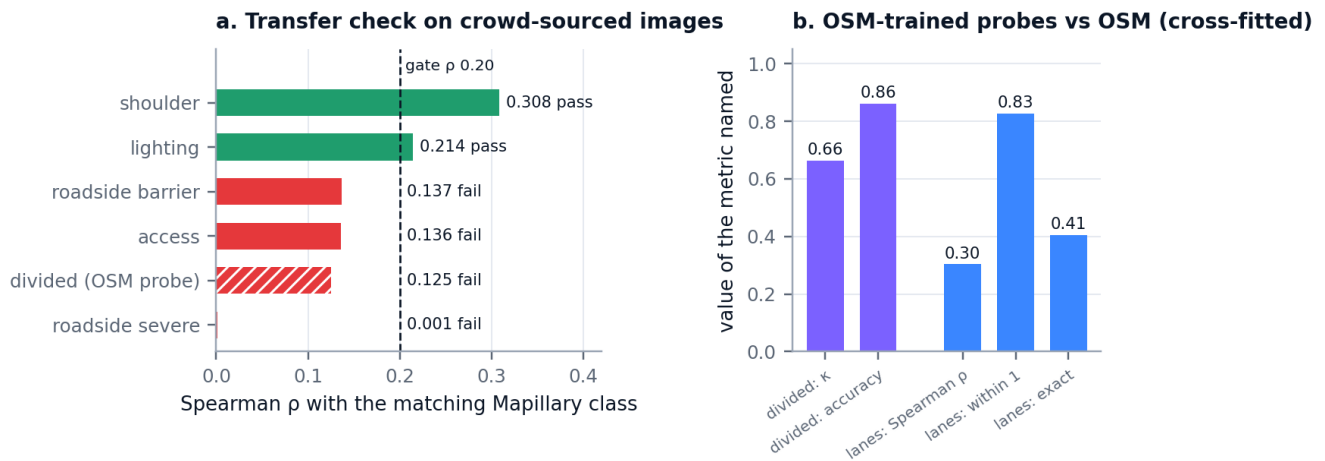


Figure 16. a: ThaiRAP-trained probes (solid) and the OSM-trained divided-road probe (hatched) against the matching Mapillary detections. b: the OSM-trained probes against OSM tags.

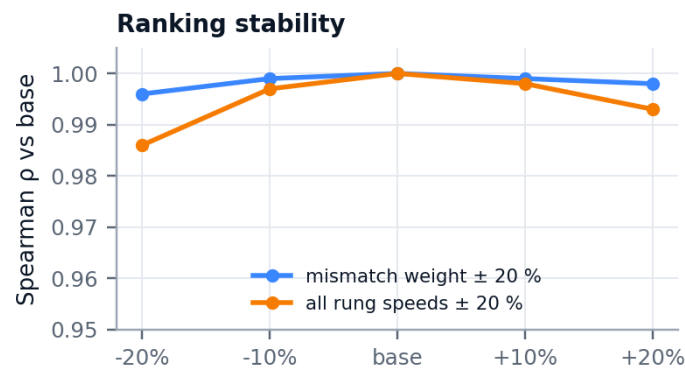


Figure 17. Rank correlation with the base ranking when the limit-gap weight or all rung speeds change by up to 20 %.

8. Results

Table 6 collects the main numbers. The score is not a speeding index: its rank correlation with the share of vehicles over the limit is 0.039, and the top 1,000 roads of the two lists barely overlap (Figure 18).

Result	Value
Segments scored	15,554 (Maharashtra 4,010, Thailand 11,544)
Road length in the High or Critical band	Maharashtra 16 %, Thailand 72 %
Harm measured from TomTom profiles	91 % of segments
Design-speed model, R^2 log v85	0.607 pooled; 0.416 MH; 0.512 TH
Vision model (Thailand), R^2 log v85	0.437 (context on the same rows 0.523)
Research ensemble, R^2 log v85	0.669 default stack; 0.665 without images; all road pieces
Research ensemble, Thai road pieces with images	0.638 without images; 0.672 with CLIP features (the package default); 0.673 with DINOv3 too
World Bank Thailand check, worst gap	2.5 pp

Table 6. Key results.

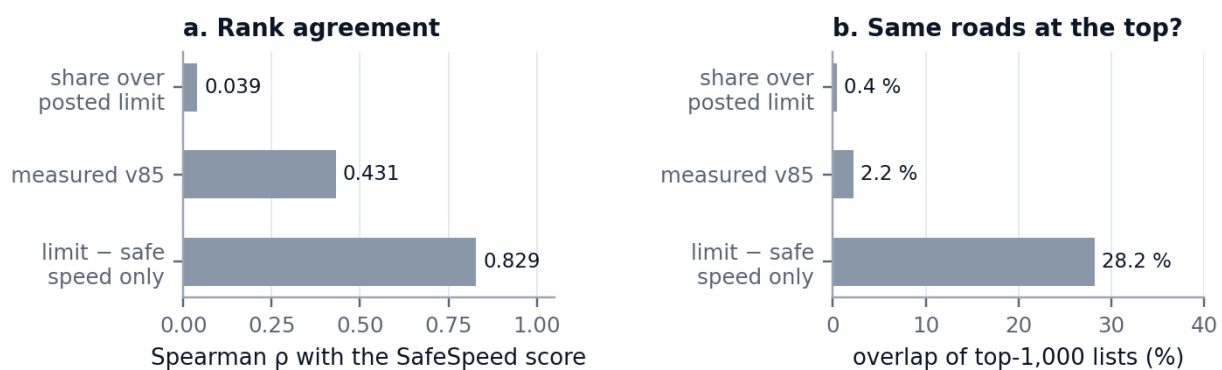


Figure 18. Rank agreement and top-1,000 overlap between the SafeSpeed score and simpler rankings.

A High or Critical band covers 72 % of the analysed road length in Thailand and 16 % in Maharashtra (Figure 19). Posted limits exceed the safe speed on 14,916 segments, and the limit gap contributes most to the score in Thailand and expected harm in Maharashtra (Figure 20).

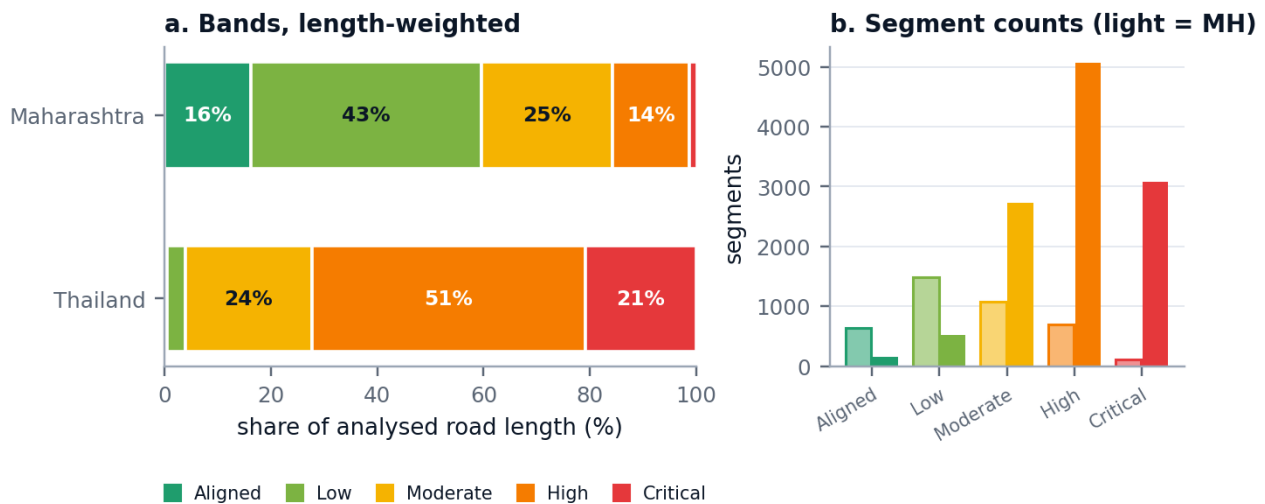


Figure 19. Bands by country. a: share of analysed road length. b: number of segments.

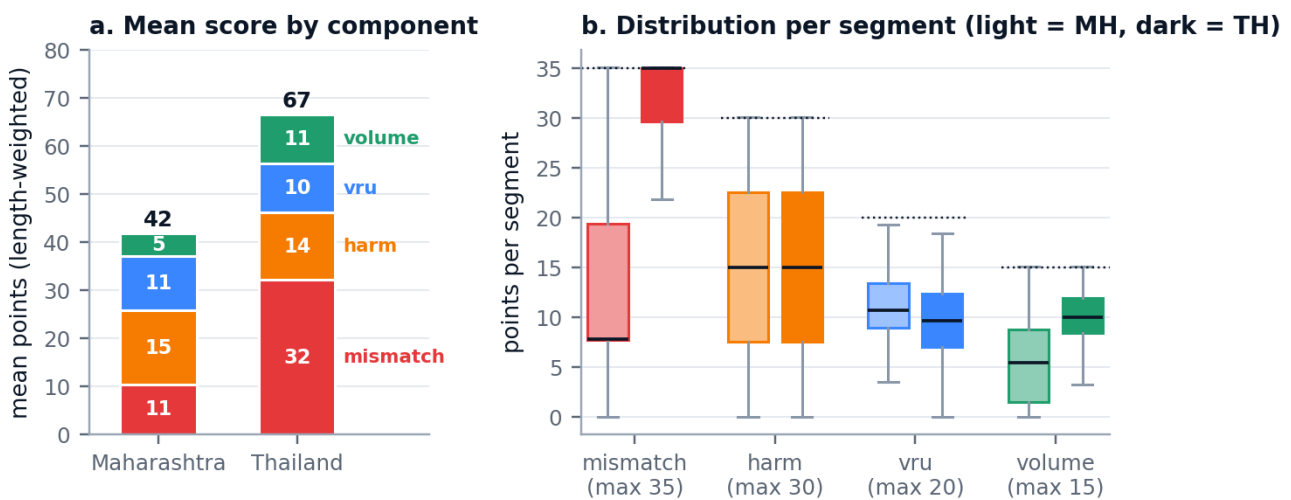


Figure 20. What drives the score. a: mean points per part, length-weighted. b: points per segment; dotted lines mark each part's maximum.

Katraj - Dehu Road Bypass (MH-3983, Maharashtra) shows the arithmetic. It is posted at 80 km/h in dense settlement, so its safe speed is 30 km/h (pedestrian impact) and it scores 96 (Figure 21).

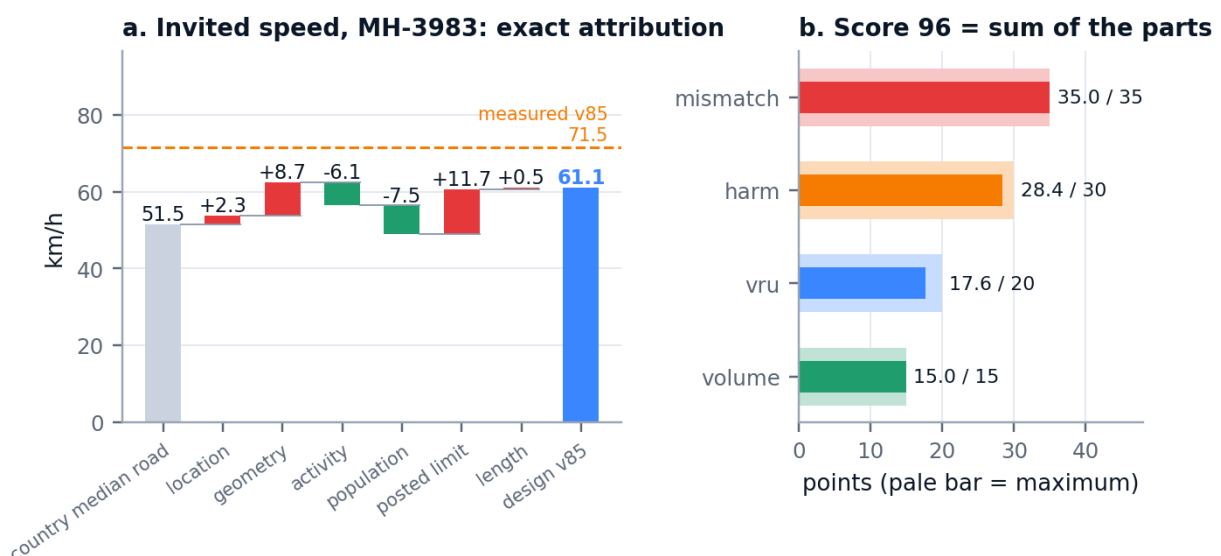


Figure 21. Worked example, MH-3983. a: the reference road plus each factor group's contribution gives the design speed exactly. b: the score is the sum of its four parts.

9. Limitations

There are no crash records, so nothing here is validated against casualties. The outputs are conditional on a crash happening, and the cost-effectiveness figure ranks actions but is not a benefit-cost ratio.

The injury curves come from German data on impacts with the front of a car, and travel speed is not impact speed. The side-impact and motorway rungs are policy values. TomTom probes are mostly cars and trucks, with no motorcycle class, and we received a single time period, so free flow and congestion are mixed.

Many posted limits are modelled rather than surveyed, and 843 were missing and imputed; the limit gap carries the largest weight. Lane counts are mostly imputed, and the semi-dense rung depends on them. Probe counts reflect fleet share as well as traffic. On a small set of well-measured roads the measured half of the harm blend uses a different crash type from the headline safe speed, which moves harm only slightly.

The speed models do not transfer between countries. The graph networks add little and react to small input changes. The image models cover Thailand only, the ThaiRAP-trained probes transfer only partly to Mapillary images, and the external agreement checks are weak (imagery audit κ 0.16; vision against headline safe speed κ 0.15).

The ThaiRAP images are licensed CC BY-NC 4.0, and the ThaiRAP label table was published without a licence. We treat everything trained on them as non-commercial: the CLIP probes, the DINOv3 heads, the vision model, the image members of the ensemble and their outputs. The V-RoAst authors' written confirmation of the labels' terms is needed before these weights are passed on. Mapillary data are CC BY-SA 4.0. The DINOv3 backbone is not shipped; it is downloaded under Meta's DINOv3 License. Whether the DINOv3 classifier heads and the cached DINOv3 features of the sample images are derivative works of DINOv3 is not settled. If they are, the DINOv3 License applies to them alongside CC BY-NC 4.0, and this has to be settled before they are passed on.

Appendix A. Variables

'ref-only' marks the columns that build the safe speed or describe who uses the road; they never enter a model. TomTom's road class (frc) and lane counts or land use read from images do enter the speed models (Section 3).

Variable	Meaning	Source	Used by
Challenge road network and TomTom aggregate speeds			
road_class, land_use	motorway to secondary; urban or rural (GRUMP extents)	RoadClass, LandUse	safe-speed fallback, VRU heuristic; ref-only
posted_limit, posted_used	posted limit, km/h; 0 → missing → (country, class) median	SpeedLimit	limit gap; model feature
median_speed, operating_speed	TomTom v50 and v85, km/h	MedianSpeed, F85thPercentileSpeed	model targets
pct_over_limit	share of vehicles over the limit	PercentOverLimit	comparison only
volume_proxy	probe count	Sample_Size_Total	volume part
length_km, lon, lat	length; first vertex	RoadLength; geometry	features, folds
helmet_rate	helmet wearing by region and land use	ADB survey constants	VRU heuristic; ref-only
Challenge feature files, joined 1:1 on the segment key			
poi_school ... poi_safety	counts of schools, hospitals, restaurants, stops, fuel and safety sites	*Count fields	VRU part; model features
poi_vru_raw	POI count weighted by relevance to people on foot	weights listed in COLUMNS.md	VRU part; model feature
seg_helmet	observed helmet use on the segment	OverallHelmetUse	reporting only; ref-only
Population, settlement, OpenStreetMap			
pop_exposure, pop_density	residents within 300 m; mean per 100 m cell	WorldPop 2025	VRU part; features
smod_rank, smod_degurba	settlement class 0-6; degree of urbanisation	GHS-SMOD R2023A	safe speed; ref-only
osm_lanes, lanes_imputed, oneway	lane count (imputed where untagged); one-way share	OSM tags + imputer	safe speed (lanes); probe labels; ref-only
Road geometry and graph			
geo_sinusosity, geo_curvature	length ÷ straight-line distance; turning per metre	linework	model features
geo_junction_density	other segment ends within 35 m, per km	network	model feature
c2g_*	graph measures: degree, PageRank, betweenness, neighbourhood size, local network form	city2graph	research ensemble

Variable	Meaning	Source	Used by
TomTom granular Traffic Stats			
P5 ... P95	19 speed percentiles, km/h	speedPercentiles	speed distribution
frc	TomTom functional road class 0-7 (0 = motorway)	TomTom granular data	research ensemble
sample_size, reliability	probe passes; n/(n + 200)	sampleSize	weights, harm credibility
harm_measured	$\int R(v) dF(v)$ for the segment's crash type	percentiles × curve	harm blend
Street images, Thailand			
v_shoulder, v_roadside_*, v_skid, v_lighting, v_calming, v_access, v_service_road	ThaiRAP attributes read by the CLIP probes	CLIP ViT-L/14 + probes	vision model; ensemble
v_lanes, v_divided	lane count; divided road	CLIP + OSM-trained probes	vision model; divided: vision safe speed only
v_ped_seen, v_moto_seen, v_sidewalk, v_crossing, ...	share of images with people, riders, footpaths, crossings	Mapillary detections	vision safe speed; ref-only
v_emb_00 ... v_emb_31, has_vision	principal components of the mean CLIP embedding; 1 if images exist	PCA	vision model; ensemble
a_<attribute>__<code>	mean probability of each iRAP code over the segment's images	DINOv3 classifier	ensemble, with_dinov3 stack only (all 19 design-role attributes, gated or not)
mly_*_imgs	images showing crossings, footpaths, lights, signals, people	Mapillary API	action checks, audit; never a feature
Outputs per segment			
safe_speed, safe_speed_crash_type_key, harm, harm_source	safe speed and its crash type; blended harm and where it comes from	ladder rule; design-speed model + TomTom	limit gap, harm part
design_speed_v50/v85, speed_attr_*, pts_*, score, band	invited speed and its split by factor group; points per part, score, band	design-speed model; score	priority
vis_*; action, action_engineering, limit_reduction_fsi_pct	vision-only speed, harm and score; recommended action, best engineering option, FSI change of a limit cut	vision model; iRAP Toolkit; Power Model	beside the score; action

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